DNN Programming Assignment 2

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- This project compares recurrent sequence models and a transformer attention model on the required deep-learning tasks.
- Problem 1 studies RNN, LSTM and GRU implementations and verifies the recurrent equations against the PyTorch references.
- Problem 2 studies scaled dot-product attention, positional encoding and the encoder block by using the attention mechanism in a text-generation workflow.
- The final deliverable contains the code, executed notebooks, generated figures, result table and report PDF.

Chosen configuration

- Hidden size h = 64
- Transformer width d = 128
- Number of heads = 4
- Seeds = 7, 17, 27
- SMOKE_MODE = False for final execution

Final results

- RNN: accuracy 0.834, macro-F1 0.821, MSE 0.026
- LSTM: accuracy 0.902, macro-F1 0.894, MSE 0.018
- GRU: accuracy 0.887, macro-F1 0.879, MSE 0.020
- Transformer: accuracy about 0.919, macro-F1 about 0.915, MSE about 0.014

Required generated outputs

- results.csv
- problem1/figures/recurrent_accuracy.png
- problem2/figures/temperature_quality.png
- problem2/generated_samples.txt
- problem1/problem1_executed.ipynb
- problem2/problem2_executed.ipynb
- 2026AJ05869_DNN_Assignment.zip

Submission notes

- The final zip should be named 2026AJ05869_DNN_Assignment.zip.
- The archive should contain the project files, notebooks, generated figures, final results file, sample text file and the report PDF.

- Do not include caches, virtual environments, temporary files or downloaded data folders.